

DOMAIN-WISE CATALOGUE

Problem Statements

69 themes and problem statements across 8 technology domains

8

DOMAINS

47

HARDWARE PROBLEMS

22

SOFTWARE PROBLEMS

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TOTAL 69 problem statements

Smart Cities & Urbanization

8 problem statements ■ 6 Hardware ■ 2 Software

No.	Theme / Title	Problem Statement	Category
1	Smart Street Environment & Noise Monitoring	Develop an IoT-based system that dynamically controls streetlight brightness based on real-time pedestrian/vehicle activity while continuously monitoring urban noise levels and identifying abnormal noise events and recurring hotspots.	HARDWARE
2	Urban Flood Monitoring & Early Warning	Develop a waterproof monitoring system that detects rapidly rising water levels in urban drains and underpasses and provides early warnings before flooding affects roads and citizens.	HARDWARE
3	Urban Underground Water Leak	Develop a system that detects and helps locate underground water-pipe leaks using flow, pressure, and acoustic sensing without requiring excavation.	HARDWARE
4	Underground Sewage Gas Safety	Develop a low-power system that continuously monitors toxic and combustible gases in underground sewage systems and provides warnings when gas concentrations reach dangerous levels.	HARDWARE
5	Dynamic Digital Traffic Signage	Develop a connected digital signage system that receives real-time traffic or road-condition data and automatically displays alternative routes or detour instructions before congested or blocked areas.	HARDWARE
6	Smart Waste Management	Develop an IoT-based waste management system that monitors garbage-bin fill levels, detects overflow conditions, and sends alerts for timely collection.	HARDWARE
7	AI-Powered Traffic Flow Optimization	Develop a centralized software platform that ingests real-time public transit and ride-sharing GPS data to dynamically adjust traffic light timings and predict congestion zones.	SOFTWARE
8	Civic Issue Crowdsourcing & Triage	Develop a web and mobile application that allows citizens to report civic issues with geotagged photos, using AI to automatically route the issue to the correct municipal department based on image classification.	SOFTWARE

Healthcare & Medical Technology

8 problem statements ■ 6 Hardware ■ 2 Software

No.	Theme / Title	Problem Statement	Category
9	Elderly Care & Assistive Technology	Develop a wearable system that detects accidental falls in elderly individuals, differentiates falls from normal movements, provides medication reminders, and automatically sends an SOS alert with the user's location to emergency contacts when a fall is detected.	HARDWARE
10	Hospital Patient Safety & Monitoring	Develop a smart monitoring system that continuously detects the remaining level of an IV fluid bag and automatically alerts nursing staff when the fluid level becomes critically low, reducing the risk of unnoticed depletion and blood backflow.	HARDWARE
11	Blood Bank & Medical Inventory Management	Develop an IoT-based system that continuously monitors blood storage temperature, tracks blood bag inventory and expiry information, and generates alerts when unsafe temperature conditions or approaching expiry dates are detected.	HARDWARE
12	Neonatal & Maternal Healthcare	Develop a smart incubator monitoring and control system that maintains stable temperature and humidity for premature babies, detects sensor or equipment failures, and provides local and remote alerts during unsafe environmental conditions.	HARDWARE
13	Preventive Healthcare & Wellness	Develop a smart bottle that automatically measures a user's water intake, monitors daily hydration patterns, and provides personalized reminders when the user needs to drink water according to predefined hydration requirements.	HARDWARE
14	AI-Based Disease Detection & Medical Diagnostics	Develop a low-cost digital microscopy system that captures blood-smear images and uses computer vision or AI to identify and highlight cells containing malaria or dengue-related abnormalities, assisting technicians in faster preliminary screening.	HARDWARE
15	Digital Healthcare & Organ Transplant Management	Develop a secure digital platform that enables hospitals to efficiently match organ donors with eligible recipients based on medically relevant compatibility factors while protecting patient identities and maintaining a transparent, tamper-resistant record of the matching process.	SOFTWARE
16	Predictive Hospital Bed Management	Develop a software solution that integrates with Electronic Health Records (EHR) to forecast patient admission rates and discharge times using machine learning, optimizing bed allocation and reducing wait times.	SOFTWARE

Electric Vehicles (EV) & Mobility

8 problem statements ■ 6 Hardware ■ 2 Software

No.	Theme / Title	Problem Statement	Category
17	EV Battery Safety & Thermal	Develop a low-cost battery monitoring and protection system that detects early signs of thermal runaway at the cell or module level and isolates the affected section while allowing the remaining battery pack to continue operating safely.	HARDWARE
18	EV Motor Fault Detection & Predictive	Develop a sensorless motor-monitoring system that analyzes three-phase current signals to detect developing inter-turn winding faults while distinguishing genuine fault signatures from normal motor imbalance and load variations.	HARDWARE
19	EV Charging Infrastructure Monitoring	Develop a retrofit device that independently verifies whether an EV charging station is actually delivering electrical power, records genuine charging events, and identifies discrepancies between reported station status and observed power delivery.	HARDWARE
20	Regenerative Braking & Energy Recovery	Develop an intelligent regenerative-braking controller that dynamically determines the appropriate level of regenerative braking based on motor conditions, battery state, temperature, and braking requirements while maximizing energy recovery and maintaining safe operation.	HARDWARE
21	EV Traction Control & Vehicle	Develop an intelligent traction-control system that detects excessive wheel slip during EV acceleration and dynamically adjusts motor torque to improve traction, vehicle stability, and energy efficiency on low-grip surfaces.	HARDWARE
22	EV Structural Health & Predictive Maintenance	Develop an accelerometer-based system for detecting structural fatigue in an EV battery mounting system by identifying changes in the mount's resonant-frequency characteristics while distinguishing actual structural degradation from normal variations in road vibration.	HARDWARE
23	Smart EV Fleet Routing & Charging	Develop a cloud-based software platform for commercial EV fleets that calculates optimal delivery routes while factoring in real-time vehicle state of charge, payload, and public/private charging station availability.	SOFTWARE
24	EV Battery Degradation Analytics	Develop a machine learning application that analyzes historical charging and usage data from electric vehicles to predict long-term battery health and recommend charging habits that extend battery lifespan.	SOFTWARE

Education & Academic Learning

7 problem statements ■ 2 Hardware ■ 5 Software

No.	Theme / Title	Problem Statement	Category
25	Multilingual Academic Learning	Develop a multilingual academic support system that accepts questions in Marathi, Hindi, or English and provides clear, simple explanations suited to different levels of student understanding while preserving the original meaning.	SOFTWARE
26	Personalized Educational Resource Recommendation	Develop a system that analyzes student interactions such as quiz response time, retry behavior, and preferred content formats to identify learning preferences and recommend educational resources suited to each student.	SOFTWARE
27	Interactive AI-Powered Quantum Learning	Develop an AI-powered interactive platform that enables users to learn quantum computing, design and simulate quantum circuits, visualize quantum states and measurement outcomes, and receive AI-based explanations, error detection, optimization suggestions, and personalized learning guidance.	SOFTWARE
28	Academic–Industry Collaboration Platform	Develop a platform that enables universities and industries to discover suitable partners, showcase their requirements and capabilities, and initiate or manage industrial visits, projects, internships, workshops, research initiatives, and other academic–industry collaborations.	SOFTWARE
29	Smart Handwriting Posture & Grip	Develop a smart pen, wearable, or similar device that detects incorrect writing posture or pencil grip in real time and provides gentle, child-friendly feedback to help students develop proper handwriting habits.	HARDWARE
30	Visual Pronunciation Learning	Develop a device that uses microphone input and a small display to provide visual mouth-shape guidance, helping language learners or speech-therapy students practice and improve pronunciation.	HARDWARE
31	AI-Driven Academic Integrity Detector	Develop a natural language processing software tool for academic institutions that analyzes student submissions to differentiate between original human writing, plagiarized text, and AI-generated content.	SOFTWARE

Renewable Energy & Power Systems

12 problem statements ■ 10 Hardware ■ 2 Software

No.	Theme / Title	Problem Statement	Category
32	Solar Energy & Battery Management	Develop an energy-management controller that monitors solar generation and load demand and intelligently schedules battery charging and discharging to reduce peak grid demand without unnecessarily exhausting the battery.	HARDWARE
33	Power Quality & Harmonic Management	Develop a real-time power-quality monitoring system that detects and analyzes harmonic components, evaluates their impact on the electrical system, and demonstrates how active compensation can improve power quality.	HARDWARE
34	Solar Microgrid & Black-Start	Develop a black-start controller that can safely restore a local renewable-energy microgrid after a complete power outage by establishing the local bus, connecting available generation and storage, and progressively restoring loads.	HARDWARE
35	Electric Vehicles & Vehicle-to-Grid Technology	Develop a smart V2G controller that coordinates EV charging and discharging according to grid demand, battery state of charge, and user requirements while maintaining EV battery availability.	HARDWARE
36	Urban Renewable Energy	Develop a small-scale energy-harvesting system that captures low-level wind or mechanical energy from urban activities such as traffic, footfall, or turbulent airflow and converts it into useful electrical energy for low-power applications.	HARDWARE
37	Railway Energy Harvesting	Develop a vibration-energy harvesting system that captures mechanical vibrations produced by railway operations and converts them into electrical energy for low-power railway monitoring applications.	HARDWARE
38	Community Microgrid & Energy Sharing	Develop an intelligent microgrid controller that detects grid failure, manages locally generated renewable energy and storage, prioritizes critical loads, and safely manages the transition between grid-connected and islanded operation.	HARDWARE
39	Wind Energy & Predictive Maintenance	Develop a wind-turbine condition-monitoring system that analyzes vibration, temperature, rotational speed, voltage, and current data to identify abnormal operating conditions before major equipment failure occurs.	HARDWARE
40	Regenerative Energy Recovery	Develop a scaled regenerative-braking system that captures mechanical energy generated during the braking of an elevator or escalator, stores the recovered energy, and measures the amount of energy recovered.	HARDWARE

No.	Theme / Title	Problem Statement	Category
41	Solar PV Predictive Maintenance & Soiling	Develop a low-cost solar-PV monitoring system that compares expected solar irradiance with actual power output to distinguish temporary reductions caused by clouds or shading from persistent problems such as dust accumulation or module degradation.	HARDWARE
42	Solar Farm Yield Forecasting	Develop a software application that integrates meteorological data and historical power output to predict daily energy generation for solar farms, enabling grid operators to balance energy loads more efficiently.	SOFTWARE
43	Microgrid Load Balancing Algorithm	Develop a software-based control system that dynamically monitors consumer energy demand within a community microgrid and autonomously redistributes available renewable energy resources to prevent localized blackouts.	SOFTWARE

Aerospace, Aviation & Space Tech

9 problem statements ■ 6 Hardware ■ 3 Software

No.	Theme / Title	Problem Statement	Category
44	UAV Safety & Autonomous Landing	Develop an autonomous emergency-landing system that monitors UAV health, detects critical failures, identifies a suitable landing zone, and guides the UAV toward a safe landing procedure.	HARDWARE
45	Autonomous Navigation & Collision	Develop an autonomous obstacle-detection and path-planning system that enables a UAV to detect obstacles in real time and modify its trajectory to maintain safe, stable, and collision-free flight.	HARDWARE
46	Energy-Efficient UAV Operations	Develop an energy-aware UAV mission-planning system that monitors battery status, distance, payload, and estimated energy consumption to optimize flight routes and automatically modify the mission when available energy becomes insufficient.	HARDWARE
47	Aircraft Electrical Systems & Fault	Develop an aircraft electrical fault-management system that monitors voltage, current, and load conditions, detects abnormal behavior, isolates faulty sections, and maintains power to critical loads through automatic network reconfiguration.	HARDWARE
48	Autonomous Search & Rescue	Develop an autonomous search-and-rescue drone system capable of systematically scanning a designated area, detecting potential survivors or targets, determining their location, and transmitting alerts to a rescue team.	HARDWARE
49	Spacecraft Power Management	Develop an intelligent satellite power-management system that monitors power generation, battery status, and load demand, and dynamically prioritizes or disconnects non-critical loads to maintain essential spacecraft operations.	HARDWARE
50	Space Safety & Debris Collision	Develop a system that tracks simulated space-debris trajectories, predicts potential intersections with a satellite's path, calculates collision risk, and determines an appropriate avoidance maneuver while minimizing energy.	SOFTWARE
51	UAV Fleet Traffic Management (UTM)	Develop a centralized software platform that coordinates flight paths for multiple commercial delivery drones in urban airspace, ensuring adherence to no-fly zones and preventing mid-air collisions.	SOFTWARE
52	Satellite Telemetry Anomaly Detection	Develop a machine learning software tool that continuously processes massive streams of satellite telemetry data to detect subtle anomalies in subsystem performance that human operators might miss.	SOFTWARE

Cybersecurity & Digital Forensics

8 problem statements ■ 4 Hardware ■ 4 Software

No.	Theme / Title	Problem Statement	Category
53	Wireless Network Security	Develop a low-cost Wi-Fi security system that detects abnormal deauthentication/disassociation activity and identifies potential rogue or duplicate access points, providing real-time alerts and maintaining a log of security events.	HARDWARE
54	USB Security & Hardware-Based Cybersecurity	Develop an inline USB security device that identifies potentially malicious or unauthorized USB devices, prevents suspicious device types from accessing the host system, provides immediate alerts, and maintains a log of USB connection events for security auditing.	HARDWARE
55	Digital Forensics & Evidence Analysis	Develop an automated forensic software tool that extracts file metadata, system artefacts, and timestamps from digital evidence, reconstructs a chronological activity timeline, generates SHA-256 hashes for evidence integrity, and produces structured investigation reports.	SOFTWARE
56	Forensic Evidence Protection	Develop a low-cost hardware write-blocking device that prevents write operations to digital storage media during forensic acquisition while allowing investigators to read and analyze the evidence, with session status and activity logging for evidence integrity verification.	HARDWARE
57	Cyber Incident Response & Digital Evidence	Develop a tamper-evident security and evidence management system that links cybersecurity alerts with collected digital evidence, maintains an immutable chain-of-custody record using cryptographic hashing, detects unauthorized modifications, and provides role-based access to investigators.	SOFTWARE
58	Secure Digital Forensics & Field Investigation	Develop a portable, access-controlled forensic evidence acquisition system that allows only authorized investigators to operate the equipment, protects digital storage from modification during acquisition, records all access attempts, and detects physical tampering with the device.	HARDWARE
59	Ransomware Behavior Isolation System	Develop endpoint security software that monitors file system activity for rapid, unauthorized encryption patterns and automatically isolates the infected machine from the corporate network before ransomware spreads.	SOFTWARE
60	Automated Phishing Threat Intelligence	Develop a software application that aggregates employee-reported suspicious emails, using domain analysis and natural language processing to verify phishing attempts and automatically update email firewall rules.	SOFTWARE

Agriculture & Aquaculture

9 problem statements ■ 7 Hardware ■ 2 Software

No.	Theme / Title	Problem Statement	Category
61	Water Management & Smart Irrigation	Over-irrigation causes water wastage and root rot, while under-irrigation reduces crop yield. Develop a smart irrigation system that monitors field conditions and controls water supply accordingly.	HARDWARE
62	Precision Agriculture – Crop Health Monitoring	Early detection of crop stress, nutrient deficiency, and disease across large fields is difficult and manual inspection is slow. Develop a UAV-based system using imaging and AI to monitor crop health and identify affected areas.	HARDWARE
63	Energy Management & Predictive Maintenance	Agricultural pumps can consume excess energy and develop faults due to overheating, overloading, abnormal current, or mechanical problems. Develop a system that monitors pump health and optimizes energy consumption.	HARDWARE
64	Climate Resilience & Crop Protection	Sudden frost events can damage or destroy crops if protective action is not taken in time. Develop a system that predicts frost conditions and automatically activates crop-protection mechanisms.	HARDWARE
65	Soil Health Monitoring (Edge AI)	Monitoring soil degradation over seasons requires complex analysis, while rural areas may have limited internet connectivity. Develop an edge-AI system that analyzes soil parameters locally and provides soil-health information.	HARDWARE
66	Aquaculture & Water Quality Management	Fish and shrimp farming require continuous monitoring of water quality, as sudden changes can cause significant losses. Develop an automated system that monitors water parameters and controls appropriate equipment when unsafe conditions occur.	HARDWARE
67	Smart Greenhouse & Environmental Monitoring	Manual monitoring of greenhouse temperature, humidity, soil conditions, and light is inefficient and can negatively affect crop growth. Develop an IoT-based system for real-time monitoring and automated environmental control.	HARDWARE
68	Crop Yield & Market Analytics Platform	Develop data analytics software that combines satellite imagery, local weather forecasts, and historical crop data to predict harvest yields and advise farmers on optimal selling times based on commodity market trends.	SOFTWARE
69	Aquaculture Feeding Optimization	Develop a software algorithm that analyzes water temperature, oxygen levels, and aquatic growth stages to calculate and schedule exact feed requirements, minimizing feed waste and preventing water pollution.	SOFTWARE